

Consolidated Course Syllabus Master Plan

June 2026

MR ROKESH TECHNOLOGY

Consolidated Course Syllabus — Master Plan
Training & Internship Programs

Philosophy: Learn → Build → Deploy → Get Hired
Methodology: 70% hands-on labs + 30% theory | Weekly mini-projects | Monthly capstones | GitHub portfolio mandatory

For duration tiers, assessment weights, and weekly rhythm see [Duration Framework](#). For package bundles see [Combo Packages](#). For project levels L1–L5 see [Project Progression](#).

Duration Framework (Applies to All Technical Tracks)

Duration	Hours	Depth	Projects	Outcome Level
2 Weeks (Crash)	40–50 hrs	Awareness + core syntax	1 mini project	Can build basic apps; needs mentorship
1 Month	80–100 hrs	Foundation + workflow	2 guided projects	Junior-ready basics
3 Months	250–300 hrs	Intermediate + integration	3–4 projects	Internship-ready
6 Months	500–600 hrs	Advanced + deployment	5–6 + 1 capstone	Job-ready (entry–mid)
9 Months (Advanced)	750–900 hrs	Expert + specialization	8–10 + 2 capstones	Industry-ready + certification prep

Weekly structure (all durations): - Mon–Thu: Concept + live coding lab - Fri: Mini-project / assignment submission - Sat: Code review + doubt clearing - Sun: Revision + documentation (GitHub README)

Assessment: Weekly quiz (20%) | Lab completion (40%) | Project (30%) | Portfolio (10%)

TRACK 1: FULL STACK DEVELOPMENT

HTML/CSS → JS/TS → React → Node/Express → Spring Boot → REST APIs → Auth → Payments → Deploy

Detailed outline: [tracks/01-full-stack-development.md](#)

2 Weeks — Crash Course

Week	Topics	Hands-On Project
W1 D1–2	HTML5 semantic tags, CSS Flexbox/Grid, responsive design	Personal landing page
W1 D3–4	JavaScript basics, DOM, events, fetch API	Interactive To-Do App
W1 D5–7	React basics: JSX, props, state, hooks	React portfolio site
W2 D1–3	Node.js + Express, REST API, CRUD	REST API for notes app
W2 D4–5	MongoDB/MySQL basics, API + DB integration	Full-stack notes app
W2 D6–7	JWT auth basics, deploy to Render/Vercel	Deploy full-stack app live

1 Month

Week	Topics	Project
1	HTML/CSS/JS + Git/GitHub	Portfolio website
2	ES6+, async JS, React fundamentals	Weather app (API)
3	React Router, forms, state management	Blog frontend
4	Node/Express, REST, MongoDB, auth, deploy	Blog CMS (MERN mini)

3 Months

Month	Topics	Projects
M1	Frontend mastery: HTML/CSS/JS/React/Tailwind	Portfolio + To-Do + Weather app
M2	Backend: Node/Express, PostgreSQL/MongoDB, REST, JWT	Task manager API + Blog CMS
M3	Full-stack integration, Spring Boot intro, payments, CI/CD basics, capstone	Capstone: E-commerce or Job Portal (MERN)

6 Months

Month	Topics	Projects
M1	HTML5, CSS3, JS ES6+, DOM, responsive design, accessibility	Portfolio + Calculator
M2	Advanced JS, async, React (hooks, context, router)	To-Do + Weather + Blog UI
M3	React advanced, Tailwind, testing (Jest), deployment	Full React SPA with routing
M4	Node.js, Express, REST APIs, error handling, file upload	REST API + Auth backend
M5	MongoDB + PostgreSQL, Mongoose, SQL joins, indexing	Database-driven app
M6	Spring Boot, JWT/OAuth, payment gateway, Docker basics, capstone	Capstone: E-commerce / Social Media Clone / Job Portal

9 Months — Advanced

Everything in 6 months **plus:** TypeScript, Next.js, microservices basics, Redis caching, WebSockets, GraphQL intro, AWS/Vercel pipelines, 2 capstones (SaaS product + Enterprise RBAC dashboard), open-source sprint, mock interviews + system design basics.

TRACK 2: DATABASE TECHNOLOGIES

SQL/MySQL → PostgreSQL → MongoDB → Firebase → Design → CRUD → API Integration → Real-time DB

Detailed outline: [tracks/02-database-technologies.md](#)

Duration	Highlights	Capstone
2 Weeks	RDBMS, MySQL, PostgreSQL, MongoDB, API integration	Connect app to DB
1 Month	SQL fundamentals through Firebase hybrid project	SQL orders + MongoDB catalog
3 Months	ER modeling, advanced SQL, MongoDB Atlas, Redis	Multi-tenant SaaS database design
6 Months	DBA, replication, Firebase, performance tuning	Hospital Management / ERP hybrid schema
9 Months	Data warehousing, ETL, migration, multi-region	Analytics platform + Real-time chat backend

TRACK 3: PROGRAMMING LANGUAGES

Python / Java / C / C++ / Dart / JavaScript + OOP + DSA + Problem Solving

Detailed outline: [tracks/03-programming-languages.md](#)

Duration	Highlights	Capstone
2 Weeks	Syntax, OOP, DSA basics (pick 1 language)	Calculator / Quiz app
1 Month	OOP, file I/O, DSA (50 LC Easy)	Mini project + 50 problems
3 Months	Language mastery, 150+ DSA problems	Custom data structure library
6 Months	Dual language (Python + Java), 300+ DSA, design patterns	Splitwise clone / Pathfinding engine
9 Months	C/C++ memory, 500+ DSA, system design	Algorithm library + Production application

TRACK 4: MOBILE APP DEVELOPMENT

React Native / Flutter → Dart/JS → UI → Navigation → API → State → Deploy

Detailed outline: [tracks/04-mobile-app-development.md](#)

Duration	Highlights	Capstone
2 Weeks	Flutter OR React Native crash	Deploy mini app
1 Month	UI, navigation, API, auth, deployment	Food ordering / Todo sync app
3 Months	Framework mastery, SQLite/Hive, push notifications	E-commerce or Food Delivery app
6 Months	Calculator → Notes → Weather → Chat → E-commerce progression	Play Store / App Store prep
9 Months	Both Flutter AND React Native, native modules, CI/CD	Marketplace + AI productivity app

TRACK 5: AI, ML & DATA SCIENCE

Python → NumPy/Pandas → Stats → ML → DL → NLP → CV → GenAI → Deployment

Detailed outline: [tracks/05-ai-ml-data-science.md](#)

Duration	Highlights	Capstone
2 Weeks	Python, Pandas, ML basics, Streamlit	Live ML dashboard
1 Month	EDA, statistics, supervised/unsupervised ML	Customer segmentation dashboard
3 Months	NLP, feature engineering, Flask/Streamlit deploy	Resume classifier / Sentiment bot
6 Months	DL (Keras), CNNs, GenAI (LangChain), MLOps	End-to-end ML pipeline with live deployment
9 Months	RNNs, CV, LLM fine-tuning, RAG, SageMaker	Multi-modal AI product + Production ML microservice

TRACK 6: CYBERSECURITY & ETHICAL HACKING

Fundamentals → Networking → Recon → Exploitation → Web Security → Reporting

Detailed outline: [tracks/06-cybersecurity-ethical-hacking.md](#)

Duration	Highlights	Capstone
2 Weeks	CIA triad, Kali, Nmap, OWASP Top 10 basics	Vulnerability assessment report
1 Month	Recon, Burp Suite, DVWA, pentest reporting	Mini pentest report
3 Months	Network security, Metasploit, CTF (10)	Full pentest report on test app
6 Months	AD attacks, cloud security, SIEM, CEH/Security+ prep	End-to-end pentest + professional report
9 Months	Red/blue team, forensics, bug bounty	Enterprise pentest + Blue team IR simulation

TRACK 7: CLOUD & SERVER TECHNOLOGIES (DevOps)

Linux → Git → Docker → Kubernetes → CI/CD → Monitoring → IaC

Detailed outline: [tracks/07-cloud-devops-server.md](#)

Duration	Highlights	Capstone
2 Weeks	Linux, Git, Docker, GitHub Actions, cloud deploy	Deploy containerized app live
1 Month	Shell scripting, CI/CD, Docker Compose, Minikube	Deploy app on Minikube
3 Months	Terraform/Ansible intro, K8s, Prometheus/Grafana	Containerized microservices on K8s
6 Months	18-lab DevOps track: GitOps, Helm, Loki, ArgoCD	Production-grade observable system
9 Months	IaC (Terraform + Pulumi), service mesh, DevSecOps	GitOps platform + Self-healing architecture

TRACK 8: CLOUD PLATFORMS (AWS / Azure / GCP)

Detailed outline: [tracks/08-cloud-platforms-aws-azure-gcp.md](#)

Duration	Highlights	Capstone
2 Weeks	AWS: IAM, EC2, S3, RDS, Lambda	Deploy 3-tier architecture
1 Month	VPC, RDS, ELB, Lambda, DynamoDB, Route 53	Full 3-tier AWS project
3 Months	AWS core + Azure + GCP fundamentals	One project per platform
6 Months	AWS SAA-C03 aligned; multi-cloud	Multi-cloud disaster recovery solution
9 Months	SAA + AZ-104 prep, ECS/EKS/AKS/GKE	Enterprise multi-tier + Serverless event platform

TRACK 9: UI/UX & WEB DESIGN

Design Thinking → Figma → Research → Wireframes → Prototypes → Handoff

Detailed outline: [tracks/09-ui-ux-web-design.md](#)

Duration	Highlights	Capstone
2 Weeks	UX/UI principles, Figma, wireframes, prototypes	Complete Figma prototype
1 Month	Research, personas, design system, case study	Portfolio case study #1
3 Months	IA, usability testing, Tailwind implementation	End-to-end UX case study + HTML/CSS build
6 Months	Designlab-aligned client projects	Real client project
9 Months	Motion design, AI-assisted design, A/B testing	SaaS product design + Enterprise design system

TRACK 10: OFFICE SUITE, TALLY & BUSINESS TOOLS

Detailed outline: [tracks/10-office-suite-tally-business-tools.md](#)

Duration	Highlights	Deliverable
2 Weeks	Word, Excel, PowerPoint, Tally Prime GST basics	Complete business report package
1 Month	Advanced Excel, Tally accounting/inventory/GST	Full company books in Tally
3–9 Months	Payroll, multi-company, ERP, client accounting projects	Live client accounting (9-month includes internship)

TRACK 11: DIGITAL MARKETING & PROMOTION

SEO → Google Ads → Social Media → Email → Analytics → Automation

Detailed outline: [tracks/11-digital-marketing-promotion.md](#)

Duration	Highlights	Capstone
2 Weeks	SEO, Google Ads, social media, GA4	Digital marketing playbook (mini)
1 Month	Full funnel: SEO audit, ads, content, email	Complete marketing playbook
3 Months	Technical SEO, paid campaigns, A/B testing	Full funnel campaign for local business
6 Months	Google Ads + Meta + HubSpot + GA4 certifications	Google Analytics Certification
9 Months	Automation (n8n/Zapier), AI content, agency management	Startup campaign + Agency portfolio (3 clients)

TRACK 12: SCHOOL & COLLEGE PROJECTS (Support Service)

Detailed outline: [tracks/12-school-college-projects.md](#)

Integrated across durations — not a standalone technical track.

Duration	Support Provided
2 Weeks	Topic selection + synopsis + basic implementation guidance
1 Month	Full mini project + documentation template
3 Months	Major project module development + weekly mentor reviews
6 Months	Complete final-year project (frontend + backend + DB) + testing
9 Months	Research-oriented project + paper writing + viva prep

TRACK 13: PROJECT REPORT / THESIS WRITING (Support Service)

Detailed outline: [tracks/13-project-report-thesis-writing.md](#)

Duration	Deliverables
2 Weeks	Synopsis + chapter outline + formatting template
1 Month	Full report structure + 2 chapters + references
3 Months	Complete report + charts/tables + plagiarism check
6 Months	Thesis-level report + slides + viva Q&A prep
9 Months	Publication-ready report + journal paper draft + defense coaching

TRACK 14: CAREER DEVELOPMENT & SOFT SKILLS

Detailed outline: [tracks/14-career-development-soft-skills.md](#)

Integrated into ALL technical tracks from Month 1.

When	Module	Activities
Every track, Week 1	Self-introduction, communication basics	Record & review intro video
1 Month+	Resume + LinkedIn	ATS-optimized resume + LinkedIn audit
3 Months+	Portfolio + GitHub	3+ projects with README case studies
6 Months+	Interview prep	Aptitude + technical mocks (weekly)
9 Months+	Placement sprint	50+ applications + mock HR/technical

RECOMMENDED COMBO PACKAGES

See full details in [00-combo-packages.md](#).

Core Packages

Package Name	Duration	Tracks Combined	Ideal For
Quick Start	2 Weeks	Any 1 track	Working professionals, skill sampling
Foundation	1 Month	Any 1 track + Career basics	Students, career switchers
Job Ready	3 Months	Full Stack OR Mobile OR AI/ML OR Cyber	Internship seekers
Professional	6 Months	Full Stack + Database + DevOps OR AI/ML + Data	Placement-focused graduates
Expert	9 Months	Full Stack + Cloud + DevOps + DSA OR AI/ML + Cloud + MLOps	Advanced career builders
Complete Tech	9 Months	All core tech tracks (staggered)	Comprehensive IT career

Department-Wise Course Map

Full categorization of all 53 tracks by academic department:

[00-department-course-map.md](#)

Department	Core Track Count	New Tracks Added
Computer Science Engineering (CSE)	9 core + electives	— (well covered)
Artificial Intelligence & Data Science	8 core	—
Information Technology (IT)	9 core	41 Network & System Admin
Electronics & Communication (ECE)	9 core	40 VLSI & Digital Design
Cyber Security	3 core	44 SOC & Digital Forensics
Artificial Intelligence & Machine Learning	6 core	43 Deep Learning & NLP
Electrical & Electronics (EEE)	5 core	39 Power Systems & Machines
Mechanical Engineering	5 core	37 CAD/CAM & Mechanical Design
Mechatronics	6 core	42 Mechatronics Integration
Civil Engineering	2 core	35 AutoCAD · 36 BIM & Structural
Fashion Technology	1 core	53 Fashion Technology

Package Name	Duration	Key Tracks
AI Professional Pack	3 / 6 / 9 mo	Prompt Eng, AI Agents, GenAI for Professionals
Generative AI for Business Pack	1 / 3 mo	Prompt Eng for Business, GenAI for Professionals
Data & BI Pack	3 / 6 mo	Data Analytics, Power BI, Tableau
DevOps Engineer Pack	3 / 6 / 9 mo	DevOps Engineering, Cloud DevOps, Cloud Platforms
Industrial Automation Pack	3 / 6 mo	PLC, Embedded Systems, Robotics & IoT
Hardware & Robotics Pack	3 / 6 / 9 mo	Arduino, Raspberry Pi, ROS2, Robotics & IoT
Drone & Autonomous Systems Pack	3 / 6 mo	Drone Programming, ROS2, Computer Vision
Embedded & IoT Pack	3 / 6 mo	Embedded Systems, Arduino, Raspberry Pi
Computer Vision Engineer Pack	3 / 6 mo	Computer Vision, AI/ML, Python
Blockchain Developer Pack	3 / 6 mo	Blockchain, Full Stack, Programming Languages
QA & Test Automation Pack	3 / 6 mo	QA Automation, Full Stack
Enterprise Business Pack	3 / 6 mo	Salesforce, SAP Basics, Data Analytics
Smart Manufacturing Pack	6 / 9 mo	PLC, ROS2, DevOps, Robotics & IoT
Fashion Designer & Brand Pack	3 / 6 mo	Fashion Technology, Digital Marketing, UI/UX

Specialized Combo Packs (Tracks 15–34)

See [00-combo-packages.md](#) for full schedules.

EXTENDED TRACKS (15–34)

Additional specialized courses. Each has a detailed outline in [tracks/](#).

#	Track	Outline	Highlights
15	DevOps Engineering	tracks/15-devops-engineering.md	SRE, GitOps, Terraform, CKA prep
16	Data Analytics	tracks/16-data-analytics.md	Excel, SQL, Python, BI dashboards
17	Prompt Engineering	tracks/17-prompt-engineering.md	RAG, LangChain, API integration (technical)
18	AI Agent Development	tracks/18-ai-agent-development.md	LangGraph, multi-agent, tool use
19	Robotics & IoT	tracks/19-robotics-iot.md	Sensors, MQTT, cloud dashboards
20	Embedded Systems	tracks/20-embedded-systems.md	C/C++, STM32, FreeRTOS
21	Raspberry Pi	tracks/21-raspberry-pi.md	GPIO, Python, edge IoT, camera
22	Arduino	tracks/22-arduino.md	Beginner hardware, sensors, Wi-Fi
23	ROS2	tracks/23-ros2.md	Navigation, SLAM, Gazebo simulation
24	Drone Programming	tracks/24-drone-programming.md	PX4, MAVLink, autonomous missions
25	PLC & Industrial Automation	tracks/25-plc-industrial-automation.md	Ladder logic, HMI, SCADA
26	Computer Vision	tracks/26-computer-vision.md	OpenCV, YOLO, deployment
27	Blockchain	tracks/27-blockchain.md	Solidity, smart contracts, dApps
28	QA Automation	tracks/28-qa-automation.md	Selenium, API testing, CI/CD
29	Power BI	tracks/29-power-bi.md	DAX, Power Query, PL-300 prep
30	Tableau	tracks/30-tableau.md	LOD, dashboards, certification prep
31	Salesforce	tracks/31-salesforce.md	Admin, Flow, Apex/LWC (9-month)
32	SAP Basics	tracks/32-sap-basics.md	MM, SD, FI modules, ERP processes
33	Prompt Engineering for Business	tracks/33-prompt-engineering-for-business.md	No-code AI for managers (non-technical)
34	Generative AI for Professionals	tracks/34-generative-ai-for-professionals.md	Multi-modal GenAI, innovation workflows
35	AutoCAD & Civil Design	tracks/35-autocad-civil-design.md	Civil Engineering

#	Track	Outline	Highlights
36	BIM & Structural Engineering	tracks/36-bim-structural-engineering.md	Civil Engineering
37	CAD/CAM & Mechanical Design	tracks/37-cad-cam-mechanical-design.md	Mechanical / Mechatronics
38	MATLAB & Simulink for Engineers	tracks/38-matlab-simulink-engineering.md	EEE / ECE / Mech / AI
39	Power Systems & Electrical Machines	tracks/39-power-systems-electrical-machines.md	EEE
40	VLSI & Digital System Design	tracks/40-vlsi-digital-system-design.md	ECE
41	Network & System Administration	tracks/41-network-system-administration.md	IT / Cyber Security
42	Mechatronics Systems Integration	tracks/42-mechatronics-systems-integration.md	Mechatronics
43	Deep Learning & NLP Specialization	tracks/43-deep-learning-nlp-specialization.md	AI & ML / AI & DS
44	SOC Analyst & Digital Forensics	tracks/44-soc-analyst-digital-forensics.md	Cyber Security

Tier Catalog Tracks (45–53)

See [00-tier-demand-catalog.md](#) for Tier 1/2/3 demand mapping.

#	Track	Outline	Tier	Department
45	CATIA Mechanical Design	tracks/45-catia-mechanical-design.md	3	Mechanical, Mechatronics
46	ANSYS Simulation & Analysis	tracks/46-ansys-simulation-analysis.md	3	Mechanical, Civil, Mechatronics, EEE
47	Electric Vehicle Technology	tracks/47-electric-vehicle-technology.md	3	EEE, Mechanical, Mechatronics, ECE
48	CNC Programming	tracks/48-cnc-programming.md	3	Mechanical, Mechatronics
49	Industrial Safety	tracks/49-industrial-safety.md	3	All Engineering
50	Cloud Computing	tracks/50-cloud-computing.md	1	CSE, IT, AI&DS, AI&ML
51	Robotics Engineering	tracks/51-robotics-engineering.md	2	Mechatronics, Mechanical, ECE
52	SolidWorks Mechanical Design	tracks/52-solidworks-mechanical-design.md	3	Mechanical, Mechatronics
53	Fashion Technology	tracks/53-fashion-technology.md	3	Fashion Technology

Recommended clusters: - **AI cluster:** 17 + 18 + 33 + 34 + 43 - **Hardware/robotics cluster:** 19 + 20 + 21 + 22 + 23 + 24 + 25 + 42 - **Data/Bi cluster:** 16 + 29 + 30 - **Enterprise cluster:** 31 + 32 + 41 - **Civil cluster:** 35 + 36 - **Mechanical/Mechatronics cluster:** 37 + 38 + 42 - **EEE cluster:** 39 + 25 + 38 - **Cyber cluster:** 06 + 41 + 44 - **Fashion cluster:** 53 + 11 + 09 + 14

See [00-department-course-map.md](#) for full branch-wise categorization (CSE, IT, ECE, EEE, Civil, Mechanical, Mechatronics, Cyber Security, AI&DS, AI&ML, **Fashion Technology**).

PROJECT PROGRESSION (Cross-Track Standard)

See full L1–L5 criteria in [00-project-progression.md](#).

Level	Example Projects	When
L1 – Mini	Calculator, To-Do, Landing page	2 weeks – 1 month
L2 – Guided	Weather app, Blog CMS, Expense tracker	1 – 3 months
L3 – Integrated	E-commerce, Job portal, Chat app	3 – 6 months
L4 – Capstone	Full production app with auth, payments, deploy	6 months
L5 – Advanced	Multi-service, cloud-native, AI-integrated platform	9 months

Sources Referenced (Trustworthy)

Source	Used For
AWS SAA-C03 Official Exam Guide	Cloud Platforms track
Ada Developers Academy Curriculum 2024	Full Stack structure
Imperial College Bootcamp Syllabus	Project-based learning model
DevOps Core Course (18 Labs)	DevOps/Cloud Server track
TCM Security – Practical Ethical Hacking	Cybersecurity labs
Designlab UX Academy	UI/UX track
TechKnowledgeHub 6-Month Curricula	Full Stack + Data Science pacing
Digital Marketing Science Syllabus	Digital Marketing track